Databases for Big Data

Exercises: Scheduling and Coordination

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Example database Sales

The following database contains two entity tables, Customer and Product, which store descriptive information about business entities. A Sales relation links these entities and records measurable events (e.g., quantities sold), forming a simple OLAP schema suitable for analytical queries.

```
CREATE TABLE Customer (
   cid      INT PRIMARY KEY,
   name     TEXT NOT NULL,
   region   TEXT
);

CREATE TABLE Product (
   pid       INT PRIMARY KEY,
   name      TEXT NOT NULL,
   category  TEXT
);

CREATE TABLE Sales (
   sale_id  INT PRIMARY KEY,
   cid      INT NOT NULL,
   pid      INT NOT NULL,
   quantity INT NOT NULL,
   FOREIGN KEY (cid) REFERENCES Customer(cid),
   FOREIGN KEY (pid)  REFERENCES Product(pid)
);
```

Here are the examples of the tables that are instances of the above relations.

Product

pid	name	category
1	Laptop	Electronics
2	Desk Chair	Furniture
3	Coffee Maker	Appliances
4	Notebook	Stationery
5	Headphones	Electronics
6	Desk Lamp	Furniture
7	Water Bottle	Accessories

Customer

cid	name	region
1	Alice Brown	West
2	Bob Smith	East
3	Carla Torres	North
4	Daniel Kim	South
5	Eva Johnson	Central

Sales

sale_id	cid	pid	quantity
1	1	1	2
2	1	4	5
3	2	3	1
4	2	7	3
5	3	2	2
6	3	5	1
7	4	1	1
8	4	6	2
9	5	3	4
10	5	7	2

Queries

Query 1.

```
select P.pid, P.category
from Product P, Sales S
where P.name = 'Notebook'
and    P.pid=S.pid
and    S.quantity>10;
```

Query 2.

```
select C.name, P.name, S.quantity
from Product P, Sales S, Customer C
where C.cid=S.cid
and    P.pid=S.pid
and    C.region='West';
```

Exercises

Exercise 1.

Describe the concepts of a query operator, an instance of an operator, a query tree as DAG of operators, a task and a task set.

Exercise 2.

Explain why the contention still happens in a lock-free algorithms for schedulling the tasks. Assume we have a multi-processor and multi-core system with NUMA memory.

Exercise 3.

Explain the concepts of a task assignment, data partitioning and data placement. In what ways are they related?

Exercise 4.

Describe the concepts of static scheduling and morsel-based scheduling as they are employed in modern OLAP DBMSs.

Exercise 5.

Sketch the sheduling algorithm used by the system Hyper.
How morsels are constructed? How the morsels are assigned to workers? What is the role of the locality of table partitions in schedulling?

Exercise 6.

Present the main concepts in the scheduling environment of Umbra.
How is a morsel defined? Why and how priorities are used in Umbra's scheduler? What the role of stride scheduling algorithm in Umbra's scheduler? What are the main elements of the Umbra's distributed scheduler?